

MAT 426, Class 08: Orthogonal Basis

Definition. A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ in $\mathbb{R}^n$ is called an **orthogonal set** if each _____ of _____ vectors in the set are _____.

Example 1. Determine whether $\left\{ \begin{bmatrix} 4 \\ 2 \\ -5 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}$ is an orthogonal set in $\mathbb{R}^3$.

Example 2. Determine whether $\left\{ \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} \right\}$ is an orthogonal set in $\mathbb{R}^3$.

THEOREM If $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent.

Proof.

Definition. An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a _____ for W that is an _____ set.

THEOREM Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$ and let $\mathbf{y}$ be any vector in W . Then, the weights $c_1, c_2, \dots, c_p$ in the linear combination

$$\mathbf{y} = c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \dots + c_p \mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}$$

Proof.

Example 3. Show that $\left\{ \begin{bmatrix} 4 \\ 2 \\ -5 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}$ form an orthogonal basis for $\mathbb{R}^3$ and express $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ as a linear combination of these vectors.

Example 4. Show that $\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -6 \\ 2 \end{bmatrix} \right\}$ form an orthogonal basis for $\mathbb{R}^2$ and express $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ as a linear combination of these vectors.

Definition. A set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ is an **orthonormal set** if it is an _____ set of _____ vectors. If W is the subspace spanned by such a set, then, $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ is an orthonormal _____ for W .

Example 5. Determine whether the given set of vectors $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is an orthonormal set. If not, normalize the vectors to form an orthonormal set.

Example 6. Determine whether the given set of vectors $\left\{ \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix} \right\}$ is orthonormal. If not, normalize the vectors to form an orthonormal set.

Vector Projection: Suppose we have a vector $\mathbf{y} \in \mathbb{R}^n$ and we want to decompose it into the sum of two vectors: one a multiple of a given direction $\mathbf{u} \in \mathbb{R}^n$ and the other vector orthogonal to $\mathbf{u}$.

Definition. The orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$ is the vector $\hat{\mathbf{y}} = \alpha \mathbf{u}$ where $\alpha = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$. The component of y that is orthogonal to $\mathbf{u}$ is the vector $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

Example 7. Find the orthogonal projection of $\mathbf{y} = \begin{bmatrix} 2 \\ 2 \\ -2 \end{bmatrix}$ onto $\mathbf{u} = \begin{bmatrix} 1/2 \\ -1/4 \\ -1/2 \end{bmatrix}$. Then write $\mathbf{y}$ as the sum of two orthogonal vectors - one in $\text{Span}\{\mathbf{u}\}$ and one orthogonal to $\mathbf{u}$.

Example 8. Find the orthogonal projection of $\begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}$ onto the line through $\begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}$ and the origin.

THEOREM An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I$.

Proof.

Example 9. Let $U = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix}$.

(a) Show that U has orthonormal columns.

(b) Verify that $\|U\mathbf{x}\| = \|\mathbf{x}\|$

THEOREM Let U be an $m \times n$ matrix with orthonormal columns, and let $\mathbf{x}, \mathbf{y}$ be in $\mathbb{R}^n$. Then,

(a) $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$

(b) $\|U\mathbf{x}\| = \|\mathbf{x}\|$

(c) $(U\mathbf{x}) \cdot (U\mathbf{y}) = 0$ if and only if $\mathbf{x} \cdot \mathbf{y} = 0$.

Proof.

Definition. An orthogonal matrix is a square invertible matrix U such that $U^T U = I$.

Example 10. Let $U = \begin{bmatrix} 3/\sqrt{11} & -1/\sqrt{6} & -1/\sqrt{66} \\ 1/\sqrt{11} & 2/\sqrt{6} & -4/\sqrt{66} \\ 1/\sqrt{11} & 1/\sqrt{6} & 7/\sqrt{66} \end{bmatrix}$.

- (a) Verify that U is an orthogonal matrix.
- (b) Show that the rows of U are also orthonormal.